



An uncommon cause of recurrent deep vein thrombosis and pulmonary embolism in a young athlete: an osteochondroma compressing the popliteal vein

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Introduction

Venous thromboembolism includes deep vein thrombosis (DVT) and pulmonary embolism, which have incidence rates of between 1 and 2 cases per 1000 people for the first episode.^{1,2} Strong, provoking risk factors for venous thromboembolism include major surgery and active cancer, but in young, otherwise healthy patients, these factors are uncommon and usually prompt a thorough search for inherited or acquired thrombophilia.³ When thrombosis recurs in the absence of identifiable risk factors, local anatomical causes should be considered.^{3,4} Osteochondroma is a benign bone tumour typically affecting children and adolescents, usually asymptomatic, but in rare cases it can produce complications by compressing adjacent arteries or veins.⁵ We present the case of an athlete, aged 20 years, with recurrent DVT and pulmonary embolism caused by a tibial osteochondroma, highlighting the

importance of considering structural causes in unexplained venous thromboembolism.

Case presentation

A previously healthy and competitive tennis player (male, aged 20 years), who also trained in weightlifting, presented with acute pain and swelling of the right calf. He had no history of trauma, immobilisation, recent travel, surgery, smoking, use of supplemental hormones for athletic activities, or family history of thrombosis. Doppler ultrasound confirmed an occlusive thrombosis of the popliteal vein with extension to the superficial femoral vein, involving the saphenopopliteal junction and the distal segment of the small saphenous vein.

Laboratory test results, including complete thrombophilia screening (antithrombin, protein C, protein S, factor V Leiden, prothrombin gene mutation, antiphospholipid antibodies, and homocysteine), were within normal range. In the absence of provoking factors, the event was classified as unprovoked. The patient was treated with rivaroxaban for 6 months (15 mg twice a day for 3 weeks, and then 20 mg once a day), and a follow-up doppler ultrasound showed no residual thrombosis or venous insufficiency, confirming complete resolution.

18 months later, the patient presented again with recurrent swelling and pain in the same leg after an intensive tennis training session. Physical examination revealed calf oedema, tenderness, and a palpable cord in the popliteal fossa. Doppler ultrasound once more showed popliteal and superficial femoral vein thrombosis. Given the patient's age and recurrence despite the absence of thrombophilia or known provoking factors, a CT venography was done 2 weeks after initial presentation to emergency care, and revealed an osteochondroma arising from the proximal tibial metaphysis, projecting posteriorly into the popliteal fossa, compressing the popliteal vein and displacing the popliteal artery (figure 1). Upper cuts through the thorax revealed signs of pulmonary embolism and a complementary chest CT angiography confirmed acute, subacute, and chronic pulmonary embolism and a small pulmonary infarct (figure 2). MRI of the right leg was done 2 months after the second DVT diagnosis, and confirmed an osteochondroma of the proximal tibia indenting both the artery and vein (figure 3).

The patient was anticoagulated with a direct oral anticoagulant for 3 months (rivaroxaban, 15 mg twice a day for 3 weeks, and then 20 mg once a day) and referred to orthopaedic surgery. Resection of the osteochondroma

Lessons for practice

- In a young, athletic patient in whom thrombophilia has been ruled out as the cause of deep vein thrombosis, local causes such as entrapment syndromes should be considered before the event is labelled as unprovoked.
- Recurrence in the same limb after ruling out residual venous insufficiency should strongly suggest a structural abnormality.
- Phlebography-CT or MRI should be considered as useful complementary studies when ultrasound does not detect abnormalities and there is a high suspicion of venous entrapment syndrome.
- Early multidisciplinary intervention, such as vascular and orthopaedic surgery, can shorten the diagnostic delay and improve outcomes.

Learning points

- Recurrent DVT in young patients without identifiable risk factors after thrombophilia and venous insufficiency are ruled out warrants investigation for local anatomical causes.
- Specific cross-sectional images (CT or MRI venography) should be considered in a DVT in a young healthy patient.
- Osteochondroma can rarely compress the popliteal vein and cause DVT and pulmonary embolism.

DVT=deep vein thrombosis.

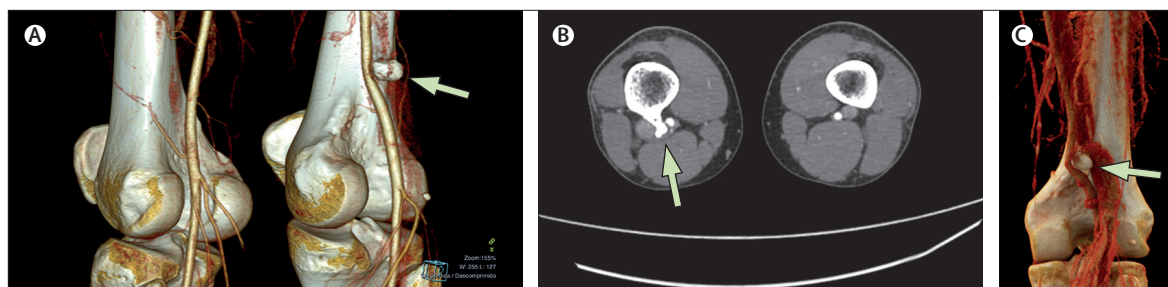


Figure 1: CT venography

There is a bony prominence from the distal third of the right femoral shaft that compressed the popliteal vein (A, B), which had an associated intraluminal hypodense band extending cranially, suggestive of subacute or chronic thrombosis. (C) The lesion also compressed the popliteal artery.

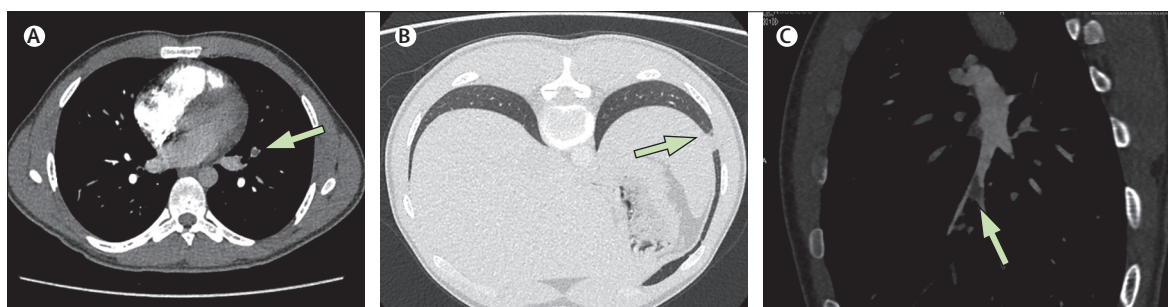


Figure 2: Chest CT angiography

(A) Acute and subacute pulmonary embolism in the lateral segmental branch of the left lower lobe. (B) A subpleural opacity suggestive of a small pulmonary infarct. (C) Chronic band-like embolism in the posterior basal segment of the right lower lobe.

was done through a posterior approach, with immediate decompression of the popliteal vein. Postoperative recovery was uneventful. Histopathological analysis confirmed the osteochondroma diagnosis. Anticoagulation was continued for 3 months postoperatively. At 1-year follow-up, the patient remained asymptomatic, with no recurrence of thrombosis or embolism, and had resumed sporting activities without restriction.

Discussion

DVT and pulmonary embolism are uncommon in young adults without recognised risk factors, and recurrence should raise suspicion of overlooked underlying causes. In such cases, the diagnostic tests usually expand to rare thrombophilias or occult malignancy.³

However, local anatomical causes might be missed if vascular imaging is limited to ultrasound. Popliteal entrapment syndromes and local tumours are rare but important causes of DVT in young athletes,^{4,5} mainly runners, cyclists, basketball players, martial artists, and tennis players as a result of repetitive knee movements, flexo-extension, and muscle and ligament hypertrophy common in these sports.

Osteochondroma, although common among benign bone tumours, seldom causes venous complications. Most cases are discovered incidentally or present with orthopaedic symptoms such as pain, deformity, or mechanical irritation. Vascular complications are rare and usually due to arterial compression (pseudoaneurysm,

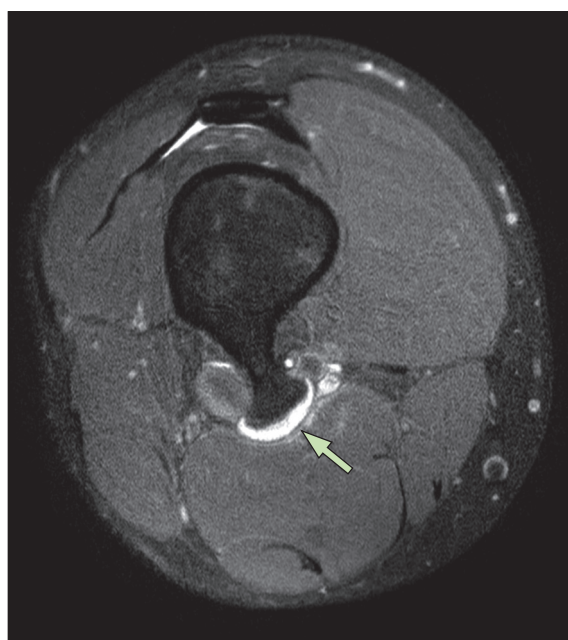


Figure 3: Right leg MRI

Focal lesion consistent with osteochondroma of the proximal right tibia, projecting into the popliteal fossa and indenting both the artery and vein.

arterial thrombosis, or ischaemia due to repetitive trauma). Venous complications are even less frequent, with only isolated cases of compression-related DVT reported.

In this case, repetitive flexion-extension during sports likely intensified venous compression, promoting thrombosis. The first episode was misclassified as idiopathic, delaying recognition of the underlying lesion. Earlier use of cross-sectional imaging might have prevented recurrence and pulmonary embolism. CT or MRI venography should be considered when unexplained or recurrent unilateral DVT occurs in young, otherwise healthy patients.

Surgical excision is the treatment of choice for symptomatic osteochondroma with vascular involvement. In our patient, resection prevented further thrombotic events and allowed discontinuation of anticoagulation, reducing the risks of bleeding and chronic venous insufficiency. However, earlier recognition of the anatomical cause could have prevented the second thrombotic event and the pulmonary embolism.

To our knowledge, this case is among the few reports of DVT with both acute and chronic pulmonary embolism caused by tibial osteochondroma. Awareness of such rare but treatable conditions can improve outcomes. This case reinforces the idea that, in young patients without

thrombophilia with recurrent unprovoked DVT, entrapment syndromes due to anatomical abnormalities should be considered among the uncommon differential diagnoses.

Declaration of interests

I declare no competing interests.

Acknowledgments

I declare that I have obtained the patient's written informed consent for the submission and publication of this manuscript.

References

- 1 Khan F, Tritschler T, Kahn SR, Rodger MA. Venous thromboembolism. *Lancet* 2021; **398**: 64–77.
- 2 Di Nisio M, van Es N, Büller HR. Deep vein thrombosis and pulmonary embolism. *Lancet* 2016; **388**: 3060–73.
- 3 Sinha S, Houghton J, Holt PJ, Thompson MM, Loftus IM, Hinchliffe RJ. Popliteal entrapment syndrome. *J Vasc Surg* 2012; **55**: 252–62.e30.
- 4 Chen CK, Kolber M. Venous popliteal entrapment syndrome. *Cardiovasc Diagn Ther* 2021; **11**: 1168–71.
- 5 Navadgi BC, Davies N, Beale AM, Arya R, Williamson DM. Deep venous thrombosis in a child: an unusual presentation of an osteochondroma. *J Pediatr Orthop* 2009; **29**: 312–14.

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